CLIMATE RISK · SOFT WHEAT

Wheat harvests and a warming climate

What the next decades mean for crop insurance in France

Angelica Bianconi · Technical assessment for AMIGO · November 2026

When the weather turns against the harvest

In 2016 the French wheat harvest fell 25% below normal, and the forecasting systems did not see it coming. As the climate warms, an insurer needs to know how often such losses may happen in the coming decades.

OUR GOAL, IN THREE STEPS



1. Learn from the past

find the weather that caused the losses of 2001-2019



2. Link weather and harvest

build and test a model that estimates the harvest from the weather



3. Look ahead

estimate how often losses occur in 2031-2050 and 2071-2090

The short answer

Today, losing more than 10% of a harvest is rare. In a warmer climate, these losses become common.

7% → **28%** → **58%**

Today | **2031-2050** | **2071-2090**

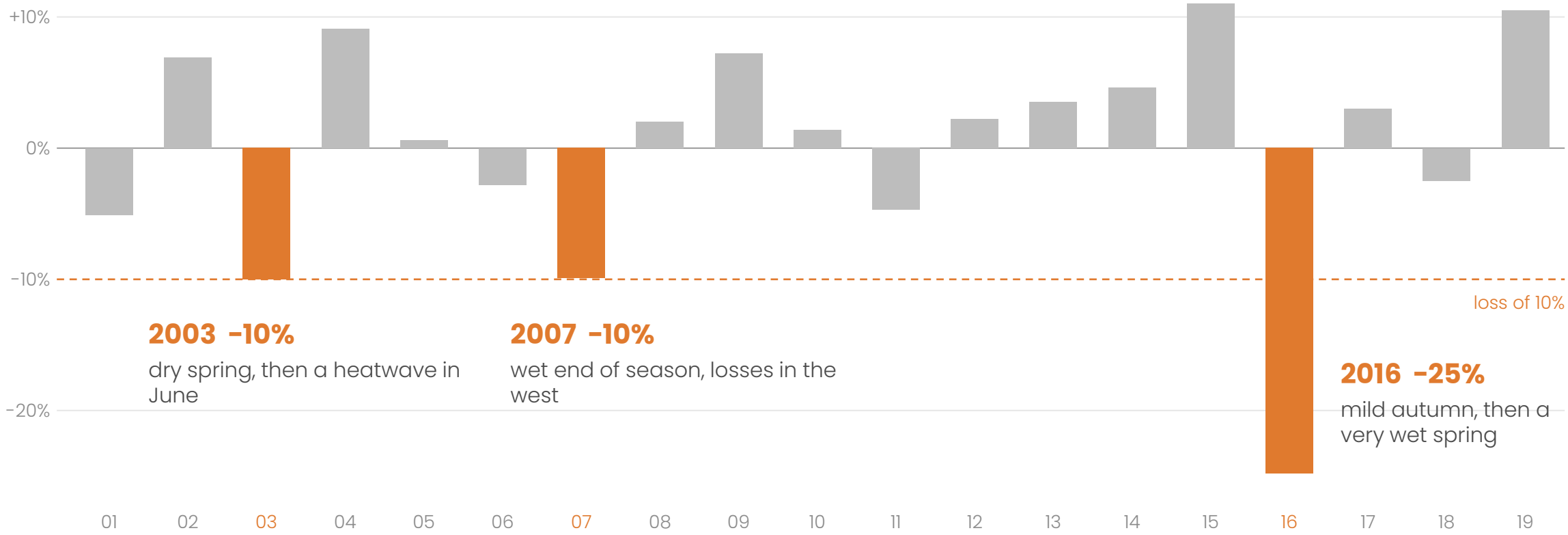
of regional harvests lose more than 10% of their expected level

The main driver is heat in May and June, when the wheat flowers.

Every French region is affected, so spreading the risk across regions helps less.

Three bad harvests in twenty years

Harvests grow over the years with better farming, so we compare each one with the level expected for its year. Most years stay within 10% of it; three seasons stand out.



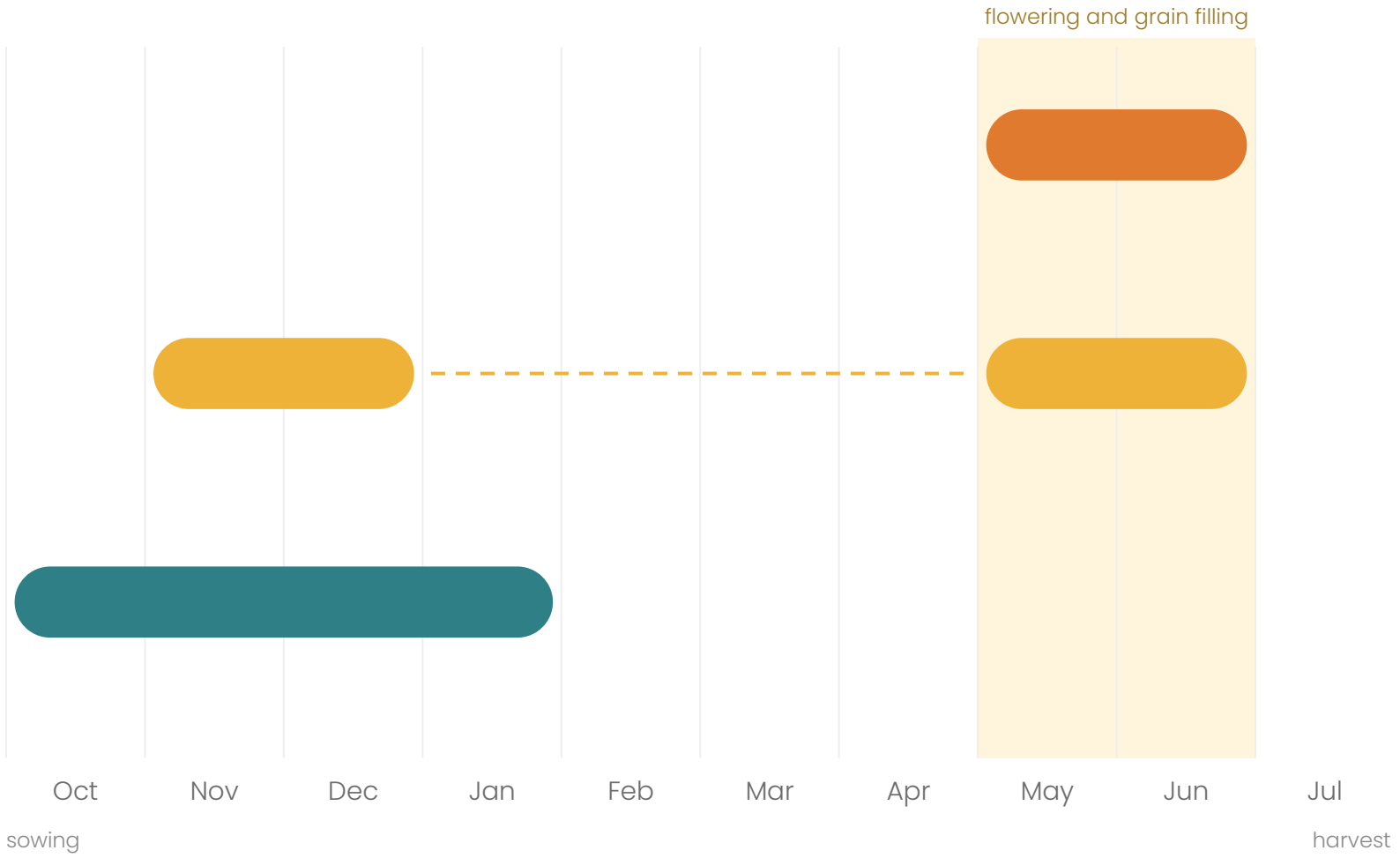
National soft wheat harvest compared with its normal level, 2001-2019

Three weather patterns explain the losses

Heat in May and June
the strongest signal, as in 2003

Mild autumn, then wet spring
harmless alone, severe together (2016)

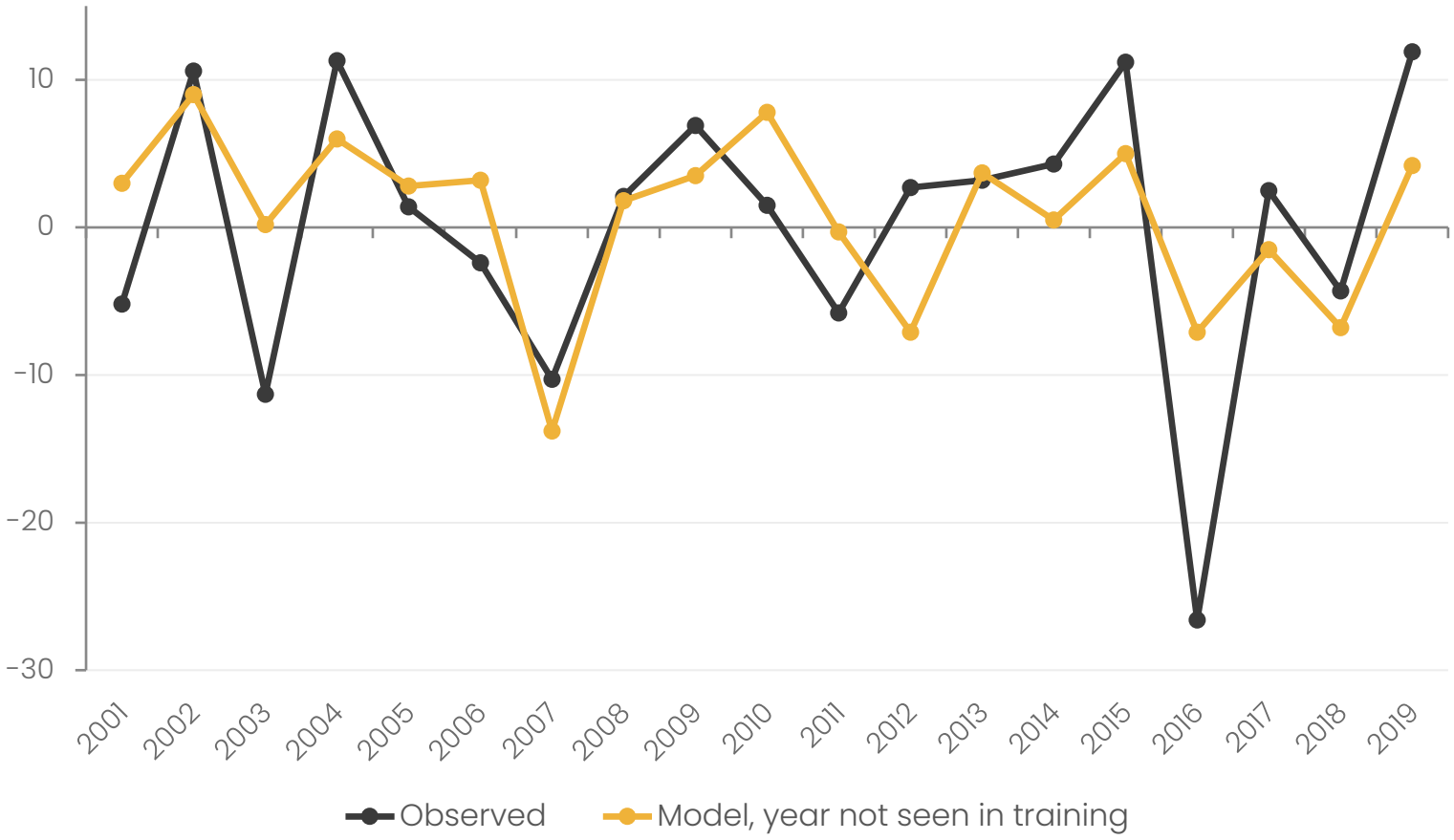
Too much rain in autumn
waterlogged fields at sowing (2001, 2007)



These three patterns became the seven weather indices used by the model.

A simple model estimates the harvest from the weather

We tested it the way it would be used: every year was predicted by a model that had never seen that year.



2007

predicted well, with a loss of 14% against 10% observed

2016

seen in the right direction, but at a quarter of its size; 2003 and 2011 missed

Few false alarms

heavy losses recognised in advance; only 5% of normal harvests are wrongly flagged as losses

Next: AI models

with more years of data, more advanced deep learning models can refine these estimates

Why the climate models need a correction

Climate models simulate the weather of the whole planet. They capture the warming, but each one is a little off on the local weather of today.



The harvest model learned from real weather

it knows how hot, cold and wet a normal season in France is



Climate models are slightly off

in France they run too warm and too wet, even in today's climate

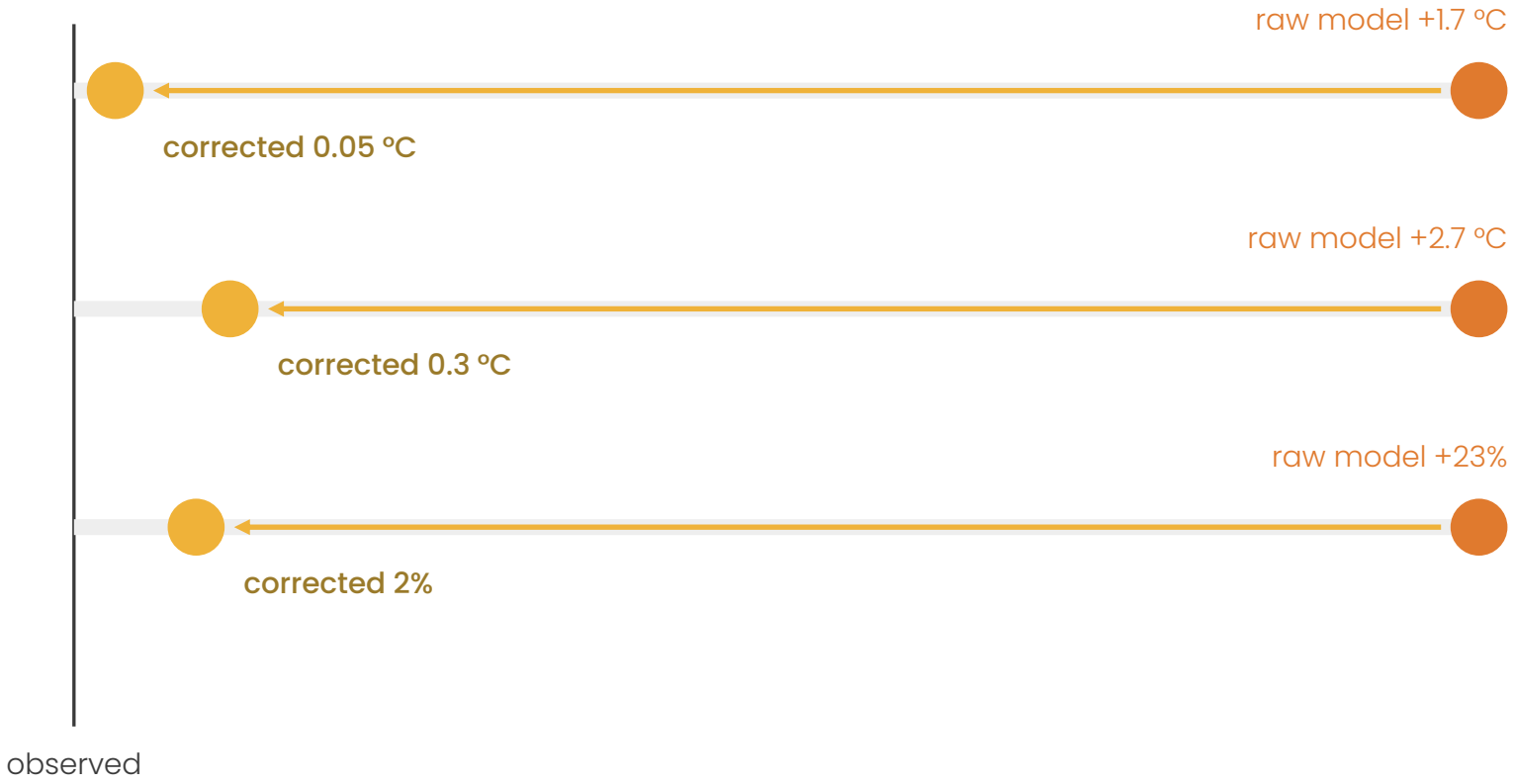


Uncorrected errors look like losses

the harvest model would read extra heat as damage that climate change did not cause

Climate models checked, then corrected

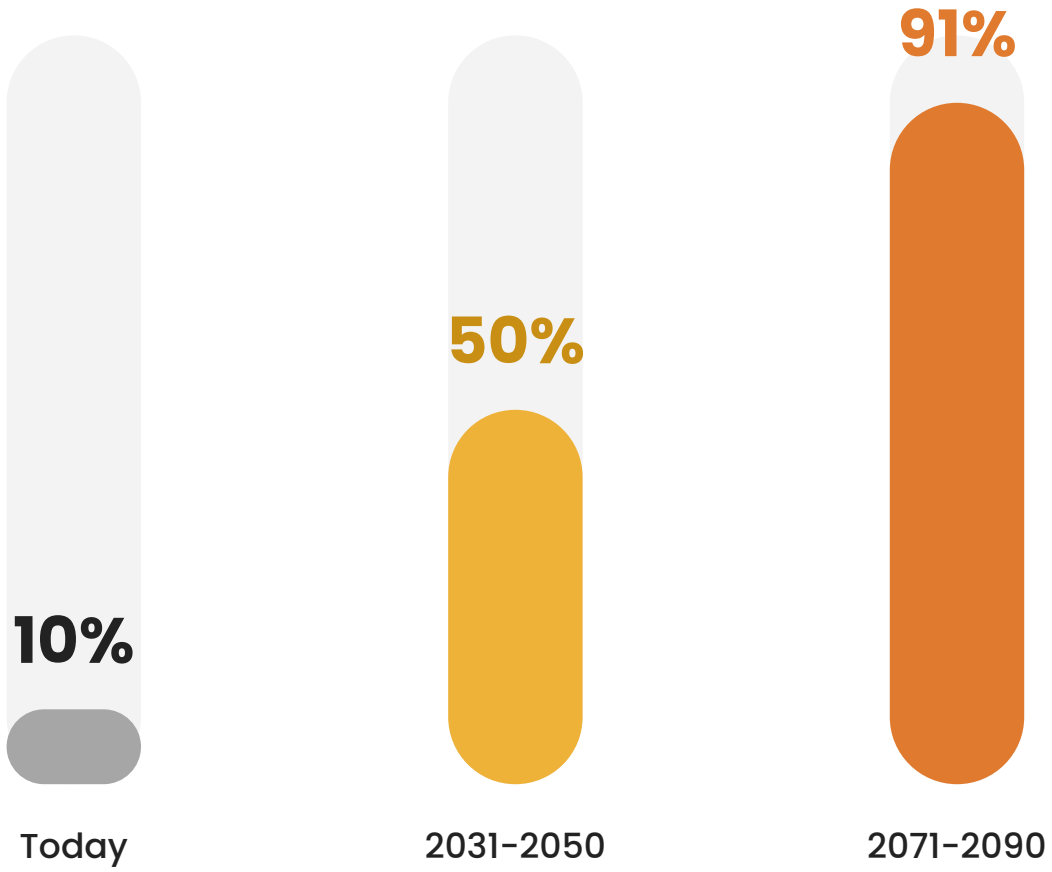
Three climate models compared with observed weather (ERA5-Land), 2000-2014



The correction removes the errors of each model and keeps the change in climate that the model simulates.

Hot late springs become the norm

We call a May-June very hot when it is hotter than 9 seasons out of 10 in 2001-2019. How often will that happen?



Seasons with a May-June as hot as 1 year in 10 today

3 → 8 → 23

days above 30 °C between May and July, per season

-20%

less rain in May and June by 2071-2090, while rain from October to January rises by about 8%

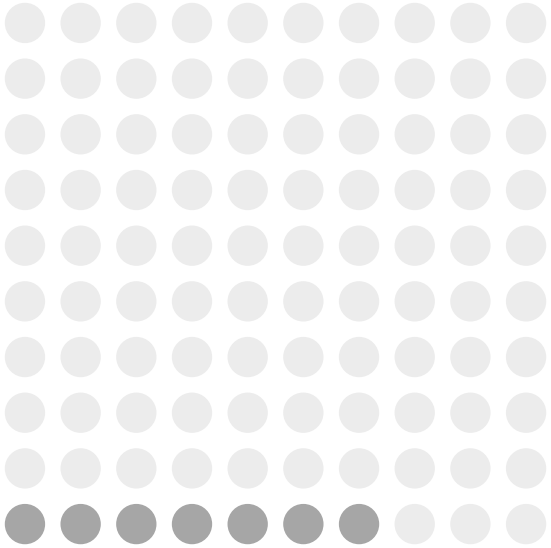
All three models

agree on the warming; they differ on how fast it comes

High-emission scenario (SSP5-8.5), the upper end of what may happen.

Heavy-loss harvests: from rare to frequent

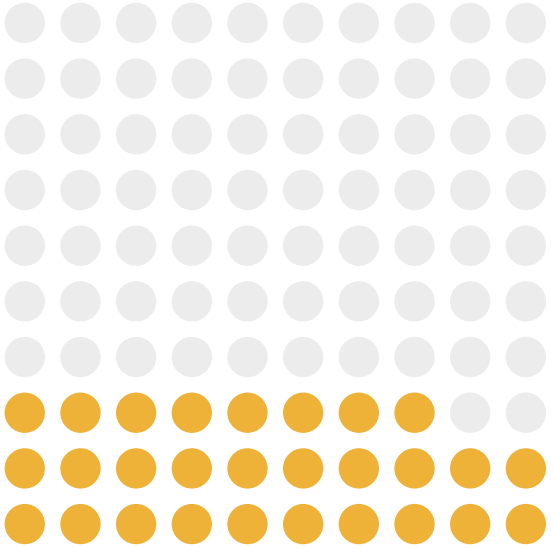
Share of regional harvests (13 regions, about 20 seasons each) with a loss larger than 10%. Each dot is 1% of the harvests.



7%

Today

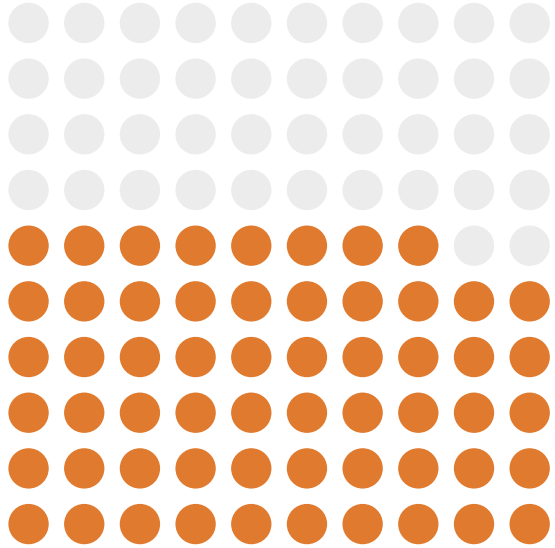
estimated by the model



28%

2031-2050

16 to 45% across climate models



58%

2071-2090

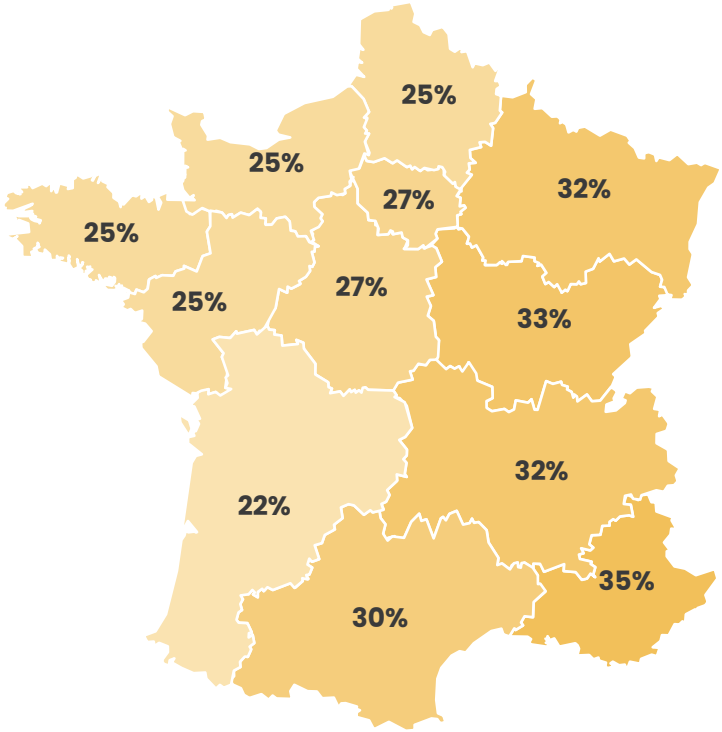
52 to 65% across climate models

On average the harvest falls 5% below today's normal by 2031-2050 and 14% by 2071-2090.

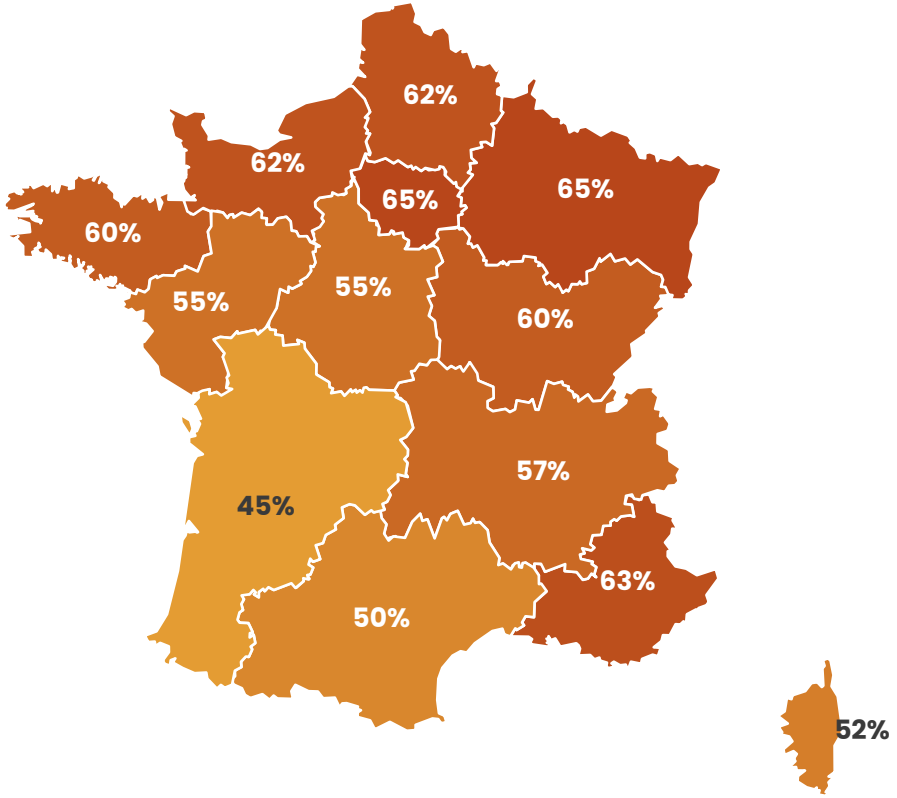
No region is spared

In every region, about a quarter of harvests lose over 10% by mid-century, and more than half by 2071-2090.

2031-2050



2071-2090



Harvests losing more than 10%

- 60%
- 50%
- 40%
- 30%
- 20%

Each regional figure comes from 20 seasons per climate model: compare the overall level, not the ranking of regions.

How sure we are, and what to do with it

How sure we are

Confident losses become more frequent, driven by heat in May and June.

Less certain how fast: mid-century estimates range from 16% to 45%.

Beyond experience by 2071-2090 most seasons are hotter than any observed: read the figure as a direction.

Assumptions high emissions and today's farming, with no adaptation.

What it means for an insurer

Premiums premiums based only on past losses will be too low, since losses become more frequent.

Early warning a hot May and June announce a poor harvest weeks before it is collected in July.

Spreading the risk a bad year in one region is less often balanced by good years elsewhere, since every region is affected.

NEXT STEPS

- Update the estimates every year with the new harvests
- Add a moderate emission scenario next to the high one
- Include adaptation in farming, and other crops